

PCAN: A Pandemic-Compatible Attentive Neural Network for Retail Sales Forecasting

Fan Li¹, Guoxuan Wang², Huiyu Chu³, Dawei Cheng^{4*} and Xiaoyang Wang¹

¹University of New South Wales, Australia

²Johns Hopkins University, USA

³Technical University of Munich, Germany

⁴Tongji University, China

{fan.li8, xiaoyang.wang1}@unsw.edu.au, gwang69@jhu.edu, lichtung191@gmail.com, dcheng@tongji.edu.cn

Abstract

The outbreak of pandemic has a huge impact on production and consumption in the business world, especially for the retail sector. As a crucial component of decision-support technology in the retail industry, sales forecasting is significant for production planning and optimizing the supply of essential goods during the pandemic. However, due to the irregular fluctuation pattern caused by uncertainty and the complex temporal correlation between multiple covariates and sales, there is still no effective approach for sales forecasting in this extreme event. To fill this gap, we propose a Pandemic-Compatible Attentive Network (PCAN) for retail sales forecasting. Specifically, to capture the irregular fluctuation patterns from the sales series, we design a fluctuation attention mechanism based on association discrepancy in the time series. Then, a parallel attention module is developed to learn the complex relationship between target sales and various dynamic influence factors in a decoupled manner. Finally, we introduce a novel rectification decoding strategy to indicate fluctuation points in prediction. By evaluating PCAN on four real-world retail food datasets from the SF Express international supply chain system, the results show that our method achieves superior performance over the existing state-of-the-art baselines. The model has been deployed in the supply chain system as a fundamental component to serve a world-leading food retailer.

1 Introduction

The global pandemics (e.g., COVID-19) would result in profound socio-economic consequences [Nicola *et al.*, 2020], especially for the retail industry [Sayyida *et al.*, 2021]. For example, the Canadian leading food retail supply chain Sysco saw a 12% drop in its sales in 2020 versus the 2019 financial year, significantly impacted by the pandemic [Ryan, 2020]. Sales forecasting plays an important role in the supply chain

management of the retail sector. It can help retailers achieve better production planning, inventory control, and financial estimation [Taşdemir, 2022], thus optimizing their business decision-making. Furthermore, accurate sales prediction can help provide the required quantities of essential goods (e.g., food and clothing) for people in need and reduce the waste of resources [Burgos and Ivanov, 2021]. This is especially important for safeguarding the well-being of people and maintaining social stability during the pandemic. However, shifts in consumer behaviors and government’s containment measures (e.g., social distancing and lockdown) can greatly alter the sales patterns of retail products, thereby disrupting the sales forecasting process [Jha *et al.*, 2023]. Thus, it is necessary yet challenging to develop a sales forecasting model that is adaptable to the pandemic scenario.

Traditional approaches like Holt-Winter’s method [Sugiarito *et al.*, 2016] and autoregressive integrated moving average (ARIMA) [Ramos *et al.*, 2015] employ techniques of time series analysis for sales forecasting. To achieve better nonlinear modeling capacity, machine learning methods such as support vector machine (SVM) [Di Pillo *et al.*, 2016] and gradient-boosting decision trees (GBDT) [Cheriyana *et al.*, 2018] are utilized and perform exceptionally well in the sales prediction task [Feizabadi, 2022]. Recently, deep learning approaches have been widely used in designing end-to-end sales forecasting models. DSF [Qi *et al.*, 2019] and MQ-RNN [Gasthaus *et al.*, 2019] are two popular deep forecasting models based on Recurrent Neural Networks (RNNs), which focus on sequential modeling. To achieve efficient parallel training and better capture long-term temporal dependencies in sales series, Convolution Neural Network (CNN) based models such as InceptionTime [Ismail Fawaz *et al.*, 2020] and TrendSpotter [Ryali *et al.*, 2023] have been devised. Furthermore, some studies attempted to apply transformer-based models, which have achieved remarkable success in Natural Language Processing, to sales forecasting [Li and Yu, 2023] (e.g., Autoformer [Zhou *et al.*, 2021a], FEDformer [Zhou *et al.*, 2022] and InParformer [Cao *et al.*, 2023]), and achieve the state-of-the-art prediction performance.

Nevertheless, developing a retail sales forecasting model adapted to the pandemic environment is challenging due to the following reasons: (1) Sales series are prone to fluctuat-

*Corresponding author

ing in an irregular manner since changes in disease transmission risks and pandemic policies can swiftly alter consumer behaviors [Beccadach *et al.*, 2020]. Existing methods fail to capture these complex fluctuation patterns in sales series, resulting in severe forecast bias. (2) Many dynamic covariates related to pandemic events (e.g., newly infected cases) and real business operations (e.g., daily promotions) may significantly influence the target sales. However, these covariate time series are often composed of intricate temporal patterns and entangled noise. How to model their latent relationship with sales and learn informative covariate representations remains challenging for existing approaches.

To address the aforementioned issues, we propose PCAN, a novel attention-based framework that is adaptable to retail sales forecasting during the pandemic period. Specifically, we first design a group embedding layer to encode heterogeneous features, including static item profiles and dynamic covariates. We also develop a series augmentation strategy to better represent target sales in the network. Next, to address the challenge of capturing abnormal fluctuation patterns, we propose a fluctuation attention module based on association discrepancy. This module effectively learns fluctuation information in sales series and guides the model to rectify its predictions accordingly. Additionally, we present a parallel attention module consisting of a pandemic attention stream (PAS) and a business attention stream (BAS), aiming at extracting sales-related dynamic features. The cross-attention mechanism within the module filters irrelevant noise in covariates and captures complex correlations between dynamic features and target sales. To make the final forecast, the decoder integrates the knowledge from the above modules and applies novel rectification coefficients learned from fluctuation information. Extensive experiments on four real-world retail food datasets from the SF Express intelligent supply chain system¹ demonstrate that PCAN consistently achieves an improvement of over 2% in ACC and a reduction of over 9% in RMSE compared to the state-of-the-art methods. The case study shows that our model significantly outperforms the existing online prediction model. The main contributions of our work can be summarized as follows:

- We present PCAN, an end-to-end pandemic-compatible attentive network, for retail sales forecasting. The model significantly enhances production planning and supply management in the retail industry during extreme events such as pandemics. It has now been deployed as a functional component in a real-world intelligent supply chain system, serving a world-leading food retailer.
- We propose a parallel stream attention module to better learn the latent relationship between dynamic covariates and sales sequence. Besides, we design a fluctuation attention mechanism to capture irregular fluctuation patterns in sales series and apply a rectification decoding strategy to inject learned fluctuation information into the final prediction.
- We conduct extensive experiments on four real-world retail food datasets and establish the new state-of-the-art

performance with a clear gain compared to the existing state-of-the-art sales forecasting methods.

Roadmap: In Section 2, we introduce the notations and formalize the problem. We present our method in Section 3 and report an empirical study in Section 4. We discuss the prior work in Section 5 and conclude our paper in Section 6.

2 Preliminary

2.1 Notations

In practice, retail sales forecasting during the pandemic is a complex temporal analysis problem with multiple influence factors. We denote historical sales of the product as $Y = \{y_t\}_{t=1}^T$, where T is the length of the time series and y_t is the sale on the day t . Moreover, we consider sales-related features X_s including static item profile P and temporal covariates C . P includes various types of time-invariant product-related inputs, such as product text attributes and store attributes. In this problem, C can be divided into C_p and C_b , which are covariate series related to pandemic events and retail business, respectively.

2.2 Problem Formulation

Given a series of historical sales Y and sales-related features $X_s = \langle P, C_p, C_b \rangle$, our task is to learn a regression model f_Θ to precisely estimate the sales of the product over period $[T + 1, T + l]$, where parameter l is the prediction horizon. The sales forecasting problem can be formulated as follows:

$$\hat{Y} = f_\Theta(Y, X_s) \quad (1)$$

where $\hat{Y} = \{\hat{y}_{T+k}\}_{k=1}^l$ are predicted sales and Θ is the set of learnable model parameters.

3 METHODOLOGY

In this section, we present the proposed PCAN in detail. We first give an overview of our architecture. After that, we describe each component of PCAN. Finally, we report the loss function and optimization strategy.

3.1 Overview

Figure 1. shows the overall architecture of PCAN. The model takes historical sales series, product profiles, pandemic-related covariates, and business-related covariates as inputs. Then, it processes each kind of feature in the group embedding layer. After that, we apply the fluctuation attention module (FAM) to detect abnormal sales fluctuations caused by pandemic events and generate instruction messages. Moreover, the parallel attention module, which consists of a pandemic attention stream (PAS) and a business attention stream (BAS), is adopted to capture complex relationships between sales and pandemic events as well as business features. Finally, the rectification decoder integrates extracted knowledge from historical sales and multiple covariates to forecast sales.

3.2 Group Embedding Layer

To better vectorize various types of input data including sales records and auxiliary features that can boost forecasting accuracy, we propose a group embedding layer with four encoding strategies: static profile embedding, temporal covariates encoding, series augmentation, and timestamp tagging.

¹<https://www.sf-international.com/>

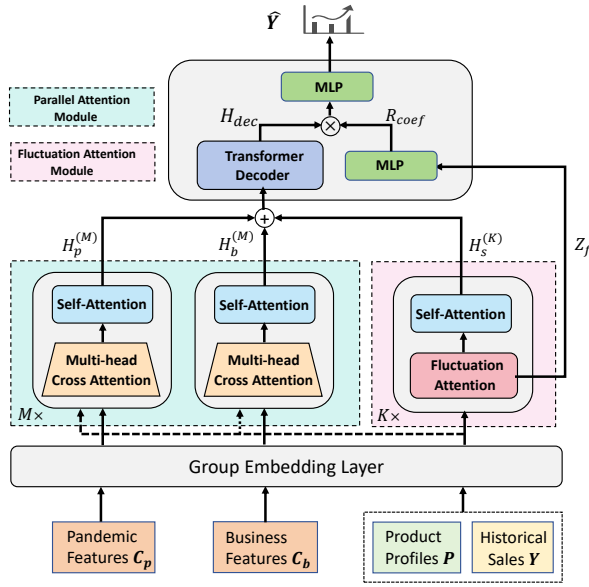


Figure 1: The pipeline of PCAN

Static Profile Embedding. The static item profile contains various time-invariant features related to the product (e.g., the product category and store location). As most of them are discrete in our task, we apply a learnable embedding method to encode them. Suppose $P = \{p_i\}_{i=1}^m$, where p_i is the i -th static feature. For each p_i , we first use one-hot encoding to transform it to a sparse vector $o_i \in \mathbb{R}^{c_i}$ where c_i is the category number of p_i . Then we map it to a column of learnable embedding matrix $\mathbf{E}_i \in \mathbb{R}^{d \times c_i}$, where d is the embedding dimension, and output dense embedding e_i as:

$$\begin{aligned} o_i &= \text{OneHot}(p_i) \\ e_i &= \mathbf{E}_i o_i \end{aligned} \quad (2)$$

We aggregate the embedded static features of each item as:

$$\mathbf{P}_s = \sum_{i=1}^m e_i \quad (3)$$

where $\mathbf{P}_s \in \mathbb{R}^d$ is the encoded static profile representation.

Temporal Covariates Encoding. For sales-related covariates, we consider pandemic events covariates $C_p \in \mathbb{R}^{N_1 \times T}$ and business covariates $C_b \in \mathbb{R}^{N_2 \times T}$, where N_1, N_2 denote the number of two covariates, respectively. We apply multi-channel 1-D convolutional filters on the time dimension to extract complex temporal information and extend zero vectors along the time dimension as placeholders for unknown future features. This process can be formulated as:

$$\begin{aligned} \mathbf{E}_p &= [\text{Conv1d}(C_p) || \mathbf{E}_p^0] \\ \mathbf{E}_b &= [\text{Conv1d}(C_b) || \mathbf{E}_b^0] \end{aligned} \quad (4)$$

where $\mathbf{E}_p, \mathbf{E}_b \in \mathbb{R}^{d \times (T+l)}$ are denoted as embedding matrices of two kinds of covariates, respectively. $\mathbf{E}_p^0, \mathbf{E}_b^0 \in \mathbb{R}^{d \times l}$ are zero padding matrices. $||$ is the concatenation operator.

Series Augmentation. The sales series $Y = \{y_t\}_{t=1}^T$ contains a single value at each time step, which cannot fully reflect the temporal pattern. We propose to augment it into a multi-variate series to better represent the status of each position in the series. These augmented sales include lag sale value, averaged sales of the last week, etc. The augmented time series $\tilde{Y}$ can be formulated as:

$$\begin{aligned} \tilde{Y} &= \{\tilde{y}_{a,1}, \dots, \tilde{y}_{a,t}, \dots, \tilde{y}_{a,T}\} \\ \tilde{y}_{a,t} &= [y_t, a_{t,1}, \dots, a_{t,i}, \dots, a_{t,N_3}] \end{aligned} \quad (5)$$

where $\tilde{y}_t \in \mathbb{R}^{N_3+1}$ is the augmented sale vector at time point t and $a_{t,i}$ refers to the i -th augmented value. N_3 denotes the number of augmentation sales. After that, we apply a linear transformation to embed this multi-value series as:

$$\mathbf{E}_s = [\mathbf{W} \tilde{Y} || \mathbf{E}_s^0] \quad (6)$$

where $\mathbf{W} \in \mathbb{R}^{d \times (N_3+1)}$ is a learnable matrix and $\mathbf{E}_s^0 \in \mathbb{R}^{d \times l}$ is the padding matrix. Finally, to inject inductive knowledge of the product attributes, we add $\mathbf{P}_s$ to each position of $\mathbf{E}_s$.

Timestamp Tagging. To make the model better capture temporal information in sales series and covariates, we utilize a time-encoding strategy. Specifically, we apply sinusoidal encoding as the local timestamp $S_l \in \mathbb{R}^{d \times (T+l)}$ to model the contextual dependencies of time and use aggregated embeddings of date features (i.e., week, month, year) as the global stamp $S_g \in \mathbb{R}^{d \times (T+l)}$ to encode date information. The embedding and aggregation methods are the same as those in static profile embedding. Then we tag the augmented sales series and two kinds of covariates with these timestamps to get model input as follows:

$$\mathbf{I}_k = \mathbf{E}_k + \mathbf{S}_l + \mathbf{S}_g \quad (7)$$

where $\mathbf{E}_k \in \{\mathbf{E}_s, \mathbf{E}_p, \mathbf{E}_b\}$.

3.3 Fluctuation Attention Module

Pandemic events will result in a chain of influences that may cause irregular fluctuations in retail sales due to their huge uncertainty. It is crucial to measure the degree of abnormal fluctuation in time series as it can help guide the model to learn more informative representations and achieve more robust forecasting performance. To quantify the degree of abnormality in time series, we apply Association Discrepancy, a time series anomaly criterion that estimates abnormality by computing the distribution discrepancy between prior association and series association. Specifically, prior association refers to the inductive bias of adjacent concentration, which implies that anomalies should primarily occur at adjacent time points that are prone to having similar abnormal patterns. The series association denotes that each time point can be characterized by its associations with all the time points, presenting as a distribution of association weights along the time horizon. This distribution offers an informative description of the temporal context. To estimate two distributions, we devise the fluctua-

tion attention function in the k -th block of the module as:

$$\begin{aligned}\mathcal{K}, \mathcal{Q}, \mathcal{V} &= \mathbf{W}^{(k)} \mathbf{H}_s^{(k-1)} \\ \mathcal{P}^{(k)} &= \text{Scale} \left(\left[\frac{1}{\sqrt{2\pi}\sigma_i} \exp\left(-\frac{|j-i|^2}{2\sigma_i^2}\right) \right]_{i,j \in \{1, \dots, T+l\}} \right) \\ \mathcal{S}^{(k)} &= \text{Softmax} \left(\frac{\mathcal{Q} \mathcal{K}^T}{\sqrt{d}} \right) \\ \mathcal{O}^{(k)} &= \mathcal{S}^{(k)} \mathcal{V}\end{aligned}\quad (8)$$

$\mathcal{K}, \mathcal{Q}, \mathcal{V} \in \mathbb{R}^{d \times (T+l)}$ corresponds to key, query, and value in self-attention. $\mathbf{H}_s^{(k-1)}$ is sales embeddings from block $k-1$ and $\mathbf{H}_s^{(0)}$ denotes sales input $\mathbf{I}_s$. $\mathbf{W}^{(k)}$ is a parameter matrix. $\mathcal{P}^{(k)}$ and $\mathcal{S}^{(k)}$ denote the prior association distribution and series association distribution, respectively. σ_i represents the i -th value of the learnable variance parameter $\sigma \in \mathbb{R}^{(T+l) \times 1}$. $\mathcal{O}^{(k)}$ is the output of the function. $\text{Scale}(\cdot)$ refers to normalization to transform the association weights to the discrete distributions $\mathcal{P}^{(k)}$. After the fluctuation attention function, we reconstruct sales representations in the k -th block as:

$$\mathbf{H}_s^{(k)} = \text{LayerNorm}(\mathcal{O}^{(k)} + \text{FFN}(\mathcal{O}^{(k)})) \quad (9)$$

where $\text{LayerNorm}(\cdot)$ denotes layer normalization and $\text{FFN}(\cdot)$ represents feed-forward neural network. We adopt KL divergence to measure association discrepancy as:

$$\text{AssDis}(\mathcal{P}^{(k)}, \mathcal{S}^{(k)}) = \text{KL}(\mathcal{P}^{(k)} \parallel \mathcal{S}^{(k)}) + \text{KL}(\mathcal{S}^{(k)} \parallel \mathcal{P}^{(k)}) \quad (10)$$

Previous work [Xu *et al.*, 2021] shows that abnormal time points will present smaller discrepancies than normal time points, which makes the criterion inherently distinguishable. In this study, we apply a multi-layer perceptron (MLP) to further extract useful messages from discrepancies, making the model pay more attention to time points that tend to have irregular fluctuations during the pandemic. This fluctuation knowledge can be computed in each layer k as:

$$\mathbf{Z}_f^{(k)} = \text{MLP}(\text{AssDis}(\mathcal{P}^{(k)}, \mathcal{S}^{(k)})) \quad (11)$$

where $\mathbf{Z}_f^{(k)} \in \mathbb{R}^{T+l}$ is the fluctuation instruction message. To aggregate the hierarchy information in the deep neural network, we concatenate the representations from each block as:

$$\mathbf{Z}_f = \parallel_{k=1}^K \mathbf{Z}_f^{(k)} \quad (12)$$

where $\mathbf{Z}_f \in \mathbb{R}^{K \times (T+l)}$ is the aggregated representation of fluctuation pattern and K is the number of stacked blocks.

3.4 Parallel Attention Stream Module

In this subsection, we design a parallel attention stream module that consists of a pandemic attention stream (PAS) and a business attention stream (BAS) to learn the latent relationship between sales and two kinds of covariates, respectively. To achieve this, we employ a two-stage attention mechanism in both submodules. In the first stage, we apply sales representations as queries to distill sales-related covariate knowledge. Particularly, we use the multi-head attention strategy,

which allows the model to jointly learn from different latent subspaces. For each type of covariate, this cross-attention mechanism in the m -th layer can be formulated as:

$$\begin{aligned}\mathcal{Q}_{s,h} &= \mathbf{W}_{s,h}^{(m)} \mathbf{I}_s \\ \mathcal{K}_{c,h}, \mathcal{V}_{c,h} &= \mathbf{W}_{c,h}^{(m)} \mathbf{H}_c^{(m-1)}, \mathbf{W}_{c,h}^{(m)} \mathbf{H}_c^{(m-1)} \\ \mathcal{O}_{c,h}^{(m)} &= \text{Softmax} \left(\frac{\mathcal{Q}_{s,h} \mathcal{K}_{c,h}^T}{\sqrt{d/H}} \right) \mathcal{V}_{c,h} \\ \mathcal{O}_c^{(m)} &= \parallel_{h=1}^H \mathcal{O}_{c,h}^{(m)}\end{aligned}\quad (13)$$

where $\mathbf{H}_c^{(m-1)}$ denotes covariate representations output from layer $m-1$. $\mathbf{W}_{c,h}^{(m)}, \mathbf{W}_{s,h}^{(m)} \in \mathbb{R}^{(d/H) \times d}$ are learnable matrices for covariate embeddings and sales embeddings, respectively. H is the number of heads. After the cross-attention function, we further stack a self-attention layer to learn context dependency within the series of covariates as:

$$\begin{aligned}\mathcal{K}_c, \mathcal{Q}_c, \mathcal{V}_c &= \mathbf{W}^{(m)} \mathcal{O}_c^{(m)} \\ \tilde{\mathcal{O}}_c^{(m)} &= \text{Softmax} \left(\frac{\mathcal{Q}_c \mathcal{K}_c^T}{\sqrt{d}} \right) \mathcal{V}_c \\ \mathbf{H}_c^{(m)} &= \text{LayerNorm}(\tilde{\mathcal{O}}_c^{(m)} + \text{FFN}(\tilde{\mathcal{O}}_c^{(m)}))\end{aligned}\quad (14)$$

where $\mathbf{H}_c^{(m)}$ is the reconstruction output of the covariates from the m -th layer. Both streams have M identical blocks.

3.5 Rectification Decoder

In this module, we propose a rectification decoding strategy, which injects the fluctuation knowledge learned from association discrepancy into decoded representations for prediction. Specifically, the output from PAS, BAS, and FAM are aggregated and then fed into a vanilla transformer decoder [Vaswani *et al.*, 2017] for the decoding process:

$$\begin{aligned}\mathbf{H}_{agg} &= \mathbf{H}_s^{(K)} + \mathbf{H}_p^{(M)} + \mathbf{H}_b^{(M)} \\ \mathbf{H}_{dec} &= \text{Transformerdecoder}(\mathbf{H}_{agg})\end{aligned}\quad (15)$$

where $\mathbf{H}_{dec} \in \mathbb{R}^{d \times (T+l)}$. To reflect fluctuation patterns in sales, we apply an MLP to learn scale factors called rectification coefficients ($\mathbf{R}_{coef}$) from $\mathbf{Z}_f$. $\mathbf{R}_{coef} \in \mathbb{R}^{T+l}$ can be viewed as unnormalized attention scores to reweigh the decoded embeddings $\mathbf{E}_{dec}$ as:

$$\begin{aligned}\mathbf{R}_{coef} &= \text{MLP}(\mathbf{Z}_f) \\ \mathbf{H}_{rec} &= \mathbf{H}_{dec} \odot \mathbf{R}_{coef}\end{aligned}\quad (16)$$

where $\mathbf{H}_{rec} \in \mathbb{R}^{d \times (T+l)}$ is the rectified output and $\odot$ denotes dot product operator. Finally, the rectified output will be forwarded to a feed-forward network and predict future sales:

$$\hat{\mathbf{Y}} = \text{FFN}(\mathbf{H}_{rec}[T+1 : T+l]) \quad (17)$$

3.6 Optimization strategy

In the training process, We take mean square error as the objective function to guide parameter learning:

$$\mathcal{L} = \frac{1}{l} \sum_{k=1}^l (y_{T+k} - \hat{y}_{T+k})^2 + \lambda \|\Theta\|^2 \quad (18)$$

Table 1: The forecasting performance comparison between PCAN and state-of-the-art baselines. The best result is bold.

	Coffee Bean			Croissant			Hot Cups			Cold Cups		
	ACC $\uparrow$	MAE $\downarrow$	RMSE $\downarrow$	ACC $\uparrow$	MAE $\downarrow$	RMSE $\downarrow$	ACC $\uparrow$	MAE $\downarrow$	RMSE $\downarrow$	ACC $\uparrow$	MAE $\downarrow$	RMSE $\downarrow$
ARIMA	53.17%	2.04	4.02	47.48%	3.02	5.71	50.15%	58.16	75.41	48.39%	62.73	90.71
Prophet	61.25%	1.49	2.21	53.08%	2.55	4.42	54.75%	40.43	59.87	58.19%	48.53	74.26
LSTNet	77.09%	0.77	1.51	59.98%	2.00	3.84	50.70%	51.02	74.57	69.25%	41.23	68.82
MQ-RNN	72.08%	0.83	1.20	60.55%	3.25	4.17	67.99%	33.30	51.64	69.56%	40.88	65.31
TCN	77.47%	0.99	1.32	72.28%	1.61	2.50	78.20%	23.17	35.04	74.70%	33.32	51.20
TrendSpotter	80.11%	0.88	1.24	73.07%	1.58	2.46	79.14%	22.73	33.45	75.48%	31.93	47.52
Informer	81.63%	0.84	1.20	69.65%	1.76	2.83	78.02%	22.82	38.50	75.82%	32.46	52.70
FEDformer	84.97%	0.64	1.03	71.18%	1.66	2.40	78.43%	22.39	33.50	76.45%	31.63	46.39
InParformer	83.75%	0.70	1.11	73.63%	1.45	2.22	79.09%	22.78	33.27	77.16%	30.65	43.41
PCAN	87.32%	0.59	1.01	76.51%	1.31	1.72	84.03%	19.38	27.38	81.22%	27.35	38.88

Table 2: The details of four products. pcs denotes unit pieces and kg represents unit kilogram.

Product	Category	Unit
Coffee Bean	raw material	kg
Croissant	food	pcs
Hot Cups	cups and lids	pcs
Cold Cups	cups and lids	pcs

where $\hat{y}_{T+k}$ is sales forecast value at the time point $T + k$, and y_{T+k} is the ground truth label in the corresponding position. λ is the regularization parameter to restrict the complexity of the deep network. We use Adam [Bello *et al.*, 2017] optimizer, with the learning rate initialized to 10^{-4} and adaptively decreased during the optimization process. Our model is implemented in PyTorch [Paszke *et al.*, 2019] and trained on an NVIDIA Tesla V100 32GB GPU.

4 EXPERIMENTS

4.1 Experimental Settings

Dataset We conduct experiments on four real-world retail food datasets from the SF Express intelligent supply chain system, which serves a world-leading food retailer. The data are collected from the retailer’s 898 offline stores in Shanghai from 01/09/2019 to 01/05/2022. During this period, the city is under the influence of the COVID-19 pandemic. The four products including Coffee Bean, Croissant, Hot Cups, and Cold Cups are among the top ten best-selling products for the retailer. We present detailed information about products in Table 2. The datasets also contain product attributes (e.g., product type and store location), pandemic features (e.g., daily number of infection cases), and business features (e.g., daily promotion type) provided by the retailer. We split the data into the train set (01/09/2019-30/04/2021), the validation set (01/05/2021-31/08/2021), and the test set (01/09/2021-01/05/2022), respectively.

Baselines To validate the effectiveness of PCAN, We compare it against the following baselines: (1) **Statistical models**: ARIMA [Box *et al.*, 2015] and Prophet [Taylor and Letham, 2018]. (2) **RNN-based models**: LSTNet [Lai *et*

al., 2018] and MQ-RNN [Gasthaus *et al.*, 2019]. (3) **CNN-based models**: TCN [Hewage *et al.*, 2020] and TrendSpotter [Ryali *et al.*, 2023]. (4) **Transformer-based models**: Informer [Zhou *et al.*, 2021a], FEDformer [Zhou *et al.*, 2022], and InParformer [Cao *et al.*, 2023].

Implementation details We set the length of historical observation T to 30 days (a month) and the prediction window l to 7 days. The dimension of feature embedding d is set to 512. We choose 8 as the number of heads in the multi-head attention mechanism. The numbers of stacked attention blocks in the parallel attention stream module, fluctuation attention module, and rectification decoder are all set to 6 in implementation. During training, the maximum epoch is set to 5, with a batch size of 128. We adopt batch normalization [Ioffe and Szegedy, 2015] with epsilon 0.01. we have released the code in <https://anonymous.4open.science/r/PCAN-33F6>

Evaluation Metrics. We employ Mean Absolute Error (MAE) and Root Mean Square Error (RMSE), which are two common time series forecasting metrics [Zhou *et al.*, 2021b]. Moreover, we use a metric utilized in the SF supply chain system called Accuracy (ACC). This metric is formulated as:

$$\text{ACC} = 1 - \text{wMAPE} = 1 - \frac{\sum_{i=1}^n |y_t - \hat{y}_t|}{\sum_{i=1}^n y_t} \quad (19)$$

where n is the prediction horizon, y_t and $\hat{y}_t$ are target and forecast sales at time step t , respectively. wMAPE refers to the Weighted Mean Absolute Percentage Error.

4.2 Performance Comparison

The overall results of different approaches are summarized in Table 1. Obviously, PCAN consistently outperforms all the baselines in all metrics across four product datasets, which validates its effectiveness for sales forecasting during the pandemic. ARIMA and Prophet perform poorly. This is because their predictions are constrained to be linear functions of past observation, and they fail to leverage multiple pandemic features. The RNN-based models (LSTNet and MQ-RNN) and CNN-based models (TCN and TrendSpotter) yield a significant improvement over these two statistical models. This strongly demonstrates the necessity of deep models for

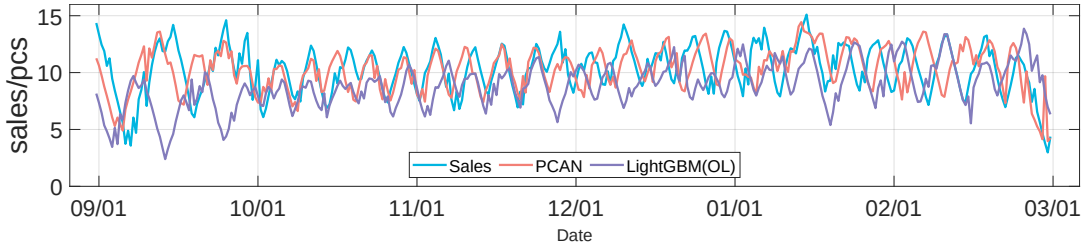


Figure 2: A case study of irregular fluctuation patterns in sales series. We report the real average sales of Croissant and prediction results from PCAN and online model LightGBM during the period of the COVID-19 pandemic in Shanghai (09/2021-03/2022).

Table 3: Ablation Analysis. PCAN w/o SA and PCAN w/o SA remove the series augmentation strategy and rectification mechanism, respectively. PCAN w/o PAS and PCAN w/o BAS remove the cross-attention mechanism in two streams, respectively. PCAN w/o FAM replaces fluctuation attention with vanilla self-attention.

Model	ACC $\uparrow$	MAE $\downarrow$	RMSE $\downarrow$
PCAN w/o SA	85.31%	0.63	1.07
PCAN w/o PAS	82.69%	0.74	1.20
PCAN w/o BAS	82.42%	0.75	1.17
PCAN w/o FAM	81.85%	0.77	1.28
PCAN w/o Rec	83.48%	0.71	1.18
PCAN	87.32%	0.59	1.01

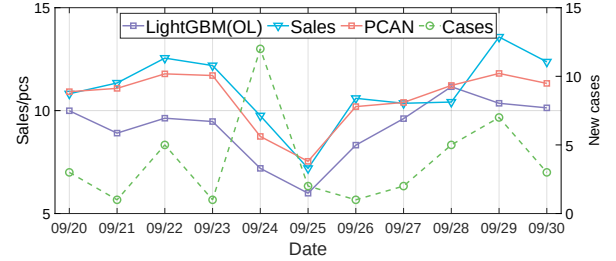


Figure 3: A case study of the influence of the pandemic on retail sales. Cases denotes the number of new COVID-19 cases each day.

sales forecasting and shows the effectiveness of deep sequential networks in extracting temporal patterns from input data. Among all baselines, transformer-based approaches (i.e., Informer, FEDformer, and InParformer) prove to be the most competitive. This is because the attention mechanism can effectively learn latent relationships between complex covariates and target sales and capture long-term patterns without decay. Notably, compared to InParformer, the state-of-the-art attention-based method, PCAN consistently achieves an improvement of over **2.8%** in terms of ACC on four product datasets. Moreover, our model yields **9.0%** ($1.11 \rightarrow 1.01$) RMSE reduction in Coffee Bean, **22.5%** ($2.22 \rightarrow 1.72$) in Croissant, **17.7%** ($33.27 \rightarrow 27.38$) in Hot Cups and **10.4%** ($43.41 \rightarrow 38.88$) in Cold Cups. The above observations demonstrate the superior performance of PCAN comprehensively. We attribute these significant and stable improvements to our specific designs. Through the parallel attention module, we can adaptively learn representations of sales-related pandemic features and business features. With fluctuation attention and rectification mechanisms, PCAN can effectively detect potential irregular fluctuations in the time series and make more accurate predictions.

4.3 Ablation Analysis

In this subsection, we verify the effectiveness of modules and techniques designed in our framework through an ablation analysis. We conduct this experiment on the Coffee Bean dataset. The experimental results are presented in Table 3. We observe that PCAN outperforms all variants across 3 metrics, demonstrating the effectiveness of each module (technique). PCAN w/o SA suffers a drop of 2.01% in ACC compared to PCAN. This proves that augmented multi-value sales

input can better represent trends in sales series. We also find a significant performance degradation when replacing the PAS or BAS with vanilla self-attention blocks. For instance, PAS brings about a 4.6% improvement in ACC and a decrease of about 0.15 in MAE. This highlights the effectiveness of our cross-attention mechanism, which utilizes sale embeddings as queries to learn sales-related covariate knowledge. Additionally, we find that PCAN w/o FAM performs poorly among several variants. The full model outperforms it by an ACC difference of 5.47%. This indicates that it is crucial to learn contextual dependency in sales series with multi-layer attention blocks. Furthermore, the fluctuation knowledge learned from the association discrepancy can also enhance model performance. To further evaluate the rectification process, we remove the R_{coef} and the scaling operation. This results in a drop of 3.84% in ACC and a 0.17 higher RMSE value. This observation further validates the effectiveness of our R_{coef} in capturing fluctuations in sales series.

4.4 Case Study

To analyze the performance of PCAN more intuitively, we conduct an empirical case study in this subsection. We compare our method with the online model (i.e. LightGBM [Ket al., 2017]) deployed in the SF Express international supply chain system to further illustrate its effectiveness in the real-world industry environment.

Case #1. In the first case, we consider all food retail stores in Shanghai from 09/2021 to 03/2022. Figure 2 shows the target average sale of Croissant in these stores and the predicted results of PCAN and the online model. We find that product sales fluctuate frequently and sharply under the impact of the pandemic. The online model fails to capture the irregular fluctuation of sales and even shows the opposite trend from the ground truth when sales fluctuate intensely. In contrast,

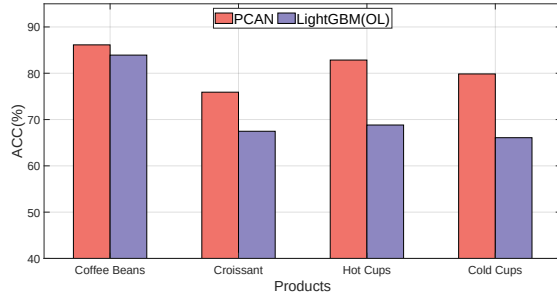


Figure 4: Performance comparison of PCAN and the online model when Shanghai is under lockdown due to the pandemic.

our PCAN reacts to abnormal fluctuations quickly and outputs accurate prediction results for a long period of time.

Case #2. In the second case, we select an offline retail store and present how the pandemic information influences product sales. We choose Croissant as the target product. In Figure 3, we plot ground truth sales, predicted sales of two models, and the daily number of new COVID-19 cases in the area around the store from 20/09/2022 to 30/09/2022. We find that when new cases increased sharply on September 24th, 2022, the sale of the product saw a drastic downward trend the next day. When new cases remained at a low level, the sales started to recover. The above observation indicates that the epidemic information has a significant impact on product sales. Moreover, we note that PCAN accurately predicts this sudden decline in sales and the subsequent recovery trend. The prediction of PCAN nearly matches the real sales during this period, while the results from the current online model show an apparent gap between target sales.

Case #3. In the third case, we compare the prediction results of PCAN and the online model in April 2022, during which Shanghai was under lockdown due to the COVID-19 pandemic. We consider all offline shops in Shanghai and report the average ACC of two models on four products. As shown in Figure 4, our model significantly outperforms the online model in terms of ACC. PCAN enhances the accuracy of Coffee Bean prediction from 83.92% to 86.13%. The performance improvements are more considerable in Croissant, which increases from 67.47% to 75.91%. Furthermore, our model demonstrates much better predictive capability for Hot Cups and Cold Cups. PCAN can achieve an ACC of over 80% on these two products, while the existing online model LightGBM can only achieve an ACC of no more than 70%. This case strongly validates the superior performance of our proposed method in the context of the pandemic.

5 RELATED WORK

5.1 Statistical Time Series Forecasting

Time-series forecasting is an active research area that covers various fields such as e-commerce, transportation, and climate [Zhang *et al.*, 2021]. Statistical models have been the most classic solutions for this problem. ARIMA [Box *et al.*, 2015] takes different ideas from autoregression (AR), moving averages (MA), and differencing (I) and combines them

to find patterns and trends in temporal data. SARIMA [Fang and Lahdelma, 2016] incorporates a seasonal moving average module, enabling the model to effectively capture seasonal trends in time series data. Prophet [Taylor and Letham, 2018] is a powerful time series forecasting framework based on an additive model that fits trends, seasonal effects, and holiday effects. Although these methods are widely utilized in real-world applications [Nigam and Shukla, 2021], most of them suffer from the strong approximation of linearity [Cui *et al.*, 2021]. Besides, they fail to consider product features and other covariates [Xin *et al.*, 2019], which are crucial for accurate sales prediction.

5.2 Learning-based Time Series Forecasting

In recent years, machine learning models have emerged as powerful tools in time series forecasting due to their flexible nonlinear modeling capacity [Masini *et al.*, 2023]. Among them, boosting regression trees such as XGBoost [Chen and Guestrin, 2016] and LightGBM [Ke *et al.*, 2017] are widely used in industry for product sales prediction [Liang *et al.*, 2019]. However, these methods require multiple handcrafted features, which can be labor-intensive. Recurrent neural networks (RNNs) such as GRU and LSTM can automatically capture temporal information in sequence data, which makes them suitable for time series prediction [Zhang *et al.*, 2021]. LSTNet [Lai *et al.*, 2018] designs a novel Recurrent-skip module to capture long-term dependence patterns and make the optimization easier. MQ-RNN [Gasthaus *et al.*, 2019] combines quantile regression with LSTM and achieves great success in the Amazon e-commerce platform. Additionally, forecasting models based on another foundational deep architecture, Convolutional Neural Networks (CNNs), have also demonstrated competitive performance. TCN [Hewage *et al.*, 2020] is a simple framework that uses 1-D convolution to model context dependencies in sequence. TrendSpotter [Ryali *et al.*, 2023] applies parallel convolution layers to learn upcoming trends in products. Lately, transformer-based models have been applied to time series prediction due to their outstanding sequential modeling capacity. Informer [Zhou *et al.*, 2021a] devises a sparse self-attention mechanism to support the long sequence forecasting task. FEDformer [Zhou *et al.*, 2022] introduces the seasonal trend decomposition method into the transformer to better capture the global view of time series. [Cao *et al.*, 2023] proposes interactive parallel attention to learn long-range dependencies in both frequency and time domains, achieving superior performance. However, the above models cannot effectively capture complex sales patterns during the pandemic.

6 Conclusion

In this work, we introduce PCAN, a pandemic-compatible attentive network for retail sales forecasting. To achieve this, we propose a fluctuation attention module based on association discrepancy in sales series to capture irregular fluctuation trends. Additionally, we develop a parallel attention module that learns intricate temporal relationships between two types of covariates and target sales series in a decoupled manner. Furthermore, we devise a rectification decoding strategy,

which predicts the volatility of sales with fluctuation patterns and heterogeneous information learned from the aforementioned modules. We conduct extensive experiments on four real-world food datasets from the SF Express supply chain system. The results demonstrate the superior performance of PCAN compared to state-of-the-art baselines. The model has been deployed as a fundamental component in the supply chain system to serve a world-leading food retailer.

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